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The acute effect of respiratory muscle training on cortisol, testosterone, and testosterone-to-cortisol ratio in well-trained triathletes - exploratory study

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ABSTRACT

The study investigated acute changes in cortisol (C) and testosterone (T) associated with a popular RMT method, voluntary isocapnic hyperpnoea (VIH), in well-trained triathletes. 19 athletes (7 females, 12 males) performed a VIH training session with pre- and post- serum C and T measurements. Repeated measures ANOVA was employed to analyze hormone changes during VIH, with additional time-sex interaction. Pearson correlation coefficient has been computed to identify the relationship between hormonal changes and age, anthropometric indices, respiratory muscle strength, and training experience. There was a statistically significant effect for C changes ($F = 13.101$, $p = 0.002$, $\eta_p^2 = 0.421$, $\omega^2 = 0.08$). The C concentration was significantly lower after VIH (Mean Difference = -32.49 ± 39.13 nmol*L^{-1}). No significant effects for T, T/C ratio, and time-sex interactions were observed ($p > 0.05$). Amongst many, significant correlations between the percentage of body fat and changes in C (r=-0.464, p=0.045), body mass and changes in T (r=0.516, p=0.024), height and changes in T (r=0.509, p=0.026) were found. VIH significantly lowered C concentration. No significant effects for T, T/C ratio, and no between-sex differences were observed. Noteworthy individual variability was observed in all the monitored indices. Significant correlations were found between acute hormone changes associated with VIH and selected anthropometric indices. The study provides initial insight into VIH's role in athletes' hormonal balance to possibly guide exercise prescription, autoregulation, arousal state management, and recovery practices in athletes.

1. Introduction

Optimal cortisol (C) and testosterone (T) levels are essential not only for achieving peak performance but also for long-term athletic improvement, and overall well-being (Casto and Edwards, 2016). Therefore, the impact of physical activity on hormonal balance has been investigated by researchers for many years now (Friedmann and Kindermann, 1989; Viru, 1992; Rudolph and McAuley, 1998). Multiple studies have analyzed acute C and T responses to different exercise modalities and protocols in various populations (Brownlee, Moore, and Hackney, 2005; Hackney et al., 2012; Fry and Lohnes, 2010; Crewther et al., 2018; Ambroży et al., 2021; Sadowska et al., 2023). However, the up-to-date investigations focused on traditional and popular exercises, whereas acute C and T responses to respiratory muscle training (RMT) remain an understudied research area.

C and T are crucial hormones that regulate exercise adaptations and metabolism (Casto and Edwards, 2016). The long-term elevated C concentration may impair performance and recovery, if not balanced by appropriate levels of anabolic hormones like T, growth hormone or insulin. That harmful hyper-cortisolism leads to the increased rate of protein breakdown known as catabolism. In contrast, T exhibits anti-catabolic and anabolic properties. The optimal T concentration is associated with increased muscle mass, strength, and improved athletic performance (Vingren et al., 2010). T/C ratio, also known as anabolic-catabolic balance, remains widely applied in sports settings to gauge individual conditioning and exercise responses (De Luccia, 2016). As health and athletic development are inextricably linked (Mackala, Michalik, and Makaruk, 2023), the optimal hormone levels are essential

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not only for peak performance but also for sustaining long-term athletic improvement and overall well-being (Cumming et al., 1986; Vingren et al., 2010; Weinberg, 2021). Noteworthy, they are also associated with the risk of injury and overtraining (Mujika et al., 1996).

RMT has been found effective in improving athletic performance for many years (Illi et al., 2012; HajGhanbari et al., 2013). The performance advantages of RMT arise from improved mechanical efficiency and increased fatigue resistance of the respiratory muscles (i.e. the diaphragm, the abdominals, and the rib cage muscles). As a result, RMT helps to delay or lessen the effects of the sympathetically driven respiratory metaboreflex (Illidi et al., 2023). The respiratory metaboreflex is a phenomenon that describes the reflex activation of the sympathetic nervous system in response to the metabolic demands of respiratory muscles. It may increase blood pressure, heart rate, and cause vasoconstriction in non-essential organs, among other effects (Harms et al., 2000). Therefore, respiratory muscle fatigue is associated with a decrease in limb blood flow, an increase in exercise-induced locomotor muscle fatigue, and is considered a performance limiter (Romer and Polkey, 2008). Improved respiratory muscle strength and endurance is expected to attenuate the respiratory metaboreflex, and minimize its systemic effects. Multiple studies have defined RMT as an ergogenic aid in endurance sports, including swimming (Kilding, Brown, and McConnell, 2010; Vašíčková, Neumannová, and Svozil, 2017; Szczepan et al., 2020), cycling (Holm, Sattler, and Fregosi, 2004; Romer, McConnell, and Jones, 2002; McMahon et al., 2002) or running (Leddy et al., 2007; Mickleborough et al., 2010). State-of-the-art literature explores the novel role of RMT techniques in warm-up, cool-down, or active recovery (Brown et al., 2008; Chiappa et al., 2009; Danek, Michalik, and Zatoń 2022; Cirino et al., 2023; Kowalski et al., 2024). Moreover, the scientific interest in the use of RMT-derived protocols in physical therapy has recently been refreshed due to its effectiveness as a post- and long-COVID rehabilitation tool (Morgan et al., 2024; Vranić et al. 2024; Calvache-Mateo et al., 2024).

VIH remains one of the most popular RMT techniques. It is a controlled breathing exercise, performed with the application of re-breathing circuits. The re-breathing circuits allow for minimal changes in blood gasometry and a lack of negative symptoms, associated with other RMT techniques (such as headache or dizziness). Typically, VIH involves breathing in and out at a moderate to high rate and volume for up to 40 minutes (McConnell, 2011). The perceived effort and physiological response to VIH are similar to low- or moderate-intensity exercise (Kowalski et al., 2023; Iqbal et al., 2023). Thus, in this regard, VIH is similar to the well-established breathing exercises applied in psychology. Numerous studies demonstrated that these exercises are at the same time associated not only with improved well-being, alleviated anxiety, and lower perceived stress (Jerath et al., 2015; Naik et al., 2018; Harbour et al., 2022), but also that they influence the C and T concentrations (Sartorio et al., 2013; Ma et al., 2017; Örün, Karaca, and Arıkan, 2021). To our knowledge, no previous studies examined the C and T changes associated with VIH.

In the current study, we provide the first, exploratory examination of the acute changes in C serum concentration, T serum concentration, and T/C ratio associated with VIH in well-trained triathletes. In doing so, we aimed to provide initial insight into VIH's role in athletes' hormonal balance to possibly guide exercise prescription, autoregulation, emotional state management, and recovery practices in sports training and physical therapy. Based on available research investigating physical activities with possibly similar intensity and duration (Davies and Few, 1973; Kaciuba-Uscilko et al., 1992; Hayes et al., 2015; Riachy, McKinney, and Tuvdendorj, 2020), we stipulated that VIH might lower C and only inconsiderably influence T serum concentrations. Moreover, the possible associations between VIH-induced hormonal changes and age, anthropometric indices, respiratory muscle strength, and training experience were explored. We hypothesized that body mass, percentage of body fat, and respiratory muscle strength are correlated with changes in C and T.

2. Methods

The study design was approved by the Institute of Sport - National Research Institute Ethics Committee (approval no KEBN-23–78-TK). Informed written consent was obtained from all study participants. All procedures were carried out following the Declaration of Helsinki.

19 well-trained triathletes (7 females, 12 males) completed the study. All were classified in Tier 3 and 4 according to the Participant Classification Framework (McKay et al., 2022), as highly-trained or elite athletes. All the participants were recruited with convenience sampling through personal connections with athletes and coaches. The criteria for study inclusion were: a valid medical certificate to participate in qualified sports, adequate performance caliber, and at least 6 years of triathlon training and racing. The exclusion criteria were: any chronic medical condition, any acute medical condition within the last 3 months, and any ongoing medication intake. Additionally, women could not use hormonal contraception for at least 3 months before the study. The required sample size was determined with G* Power (version 3.1.9.2; Germany) and totaled 16 participants. The following input parameters were set: effect size $f = 0.5$, significance $\alpha = 0.05$, power $(1 - \beta) = 0.95$, number of groups = 2, number of measurements = 2 (ANOVA with repeated measures, within-between interaction). The characteristics of participants who completed the study are presented in Table 1.

The body composition was assessed with InBody 770 (InBodyUSA, Cerritos, CA, United States). The participants' height was measured with the free-standing digital stadiometer seca 274 (seca GmbH & Co. KG., Hamburg, Germany). The S-Index Tests to assess respiratory muscle strength were performed according to the guidelines presented by Kowalski and Klusiewicz, 2023 (Kowalski and Klusiewicz, 2023) with POWERbreathe K5 (POWERbreathe International Ltd., Southam, United Kingdom).

All the participants performed one supervised VIH session and three unsupervised VIH sessions during 10 days before the testing sessions in the laboratory. The testing sessions were held between 8:30 and 10:00 AM to minimize diurnal variations of hormone concentration. The initial familiarization and the testing sessions were performed at the Institute of Sport - National Research Institute in Warsaw, Poland. The laboratory temperatures ranged from 20.2°C to 22.1°C and humidity levels between 41 % and 46 %. The participants remained seated for 15 minutes before and during the whole experimental procedure. The blood samples were taken 5' before and 5' after VIH sessions. The session consisted of 15 minutes of dynamic and vigorous breathing with a frequency of 24 breaths·min-1. The BreathWayBetter devices (Isocapnic Technologies Inc, Kelowna, Canada) with 6-liter bags were applied. The participants were instructed to employ diaphragmatic breathing patterns and limit upper chest and shoulder movements.

Roche Cobas analyzer (Roche Diagnostics at F. Hoffmann-La Roche Ltd., Basel, Switzerland) was used to measure T and C with (enhanced chemiluminescence immunoassay) (ECLIA) method for assay analysis. The blood samples were taken from the fingertip before and after the monitored RMT sessions by skilled and experienced technicians. The capillary tubes of 300 μL were used. The samples for hormone measurements underwent centrifugation at 5000 rpm for 10 minutes at a temperature of 4°C. Then, the serum was aliquoted and stored at −20°C

Table 1
Basic characteristics of study participants.

Variable/Group	Females (n=7)	Males (n=12)
Age (years)	29.8 ± 6.8	30.3 ± 4.9
Body mass (kg)	57.3 ± 1.8	75.5 ± 5.4
Body fat (%)	12.4 ± 0.4	12.2 ± 0.2
Body height (cm)	166.0 ± 4.2	183.7 ± 4.6
S-Index Test score (cmH_2O)	124.5 ± 24.1	182.1 ± 17.8
Training experience (years)	12.9 ± 4.3	12.2 ± 4.4

Values are mean ± standard deviation.

until the assays were performed during the next 24 h. The assays were performed in duplicates. The mean between-assay and within-assay errors of assessments were under 5 % for both T and C. T/C ratio was calculated as follows:

T/C ratio = (testosterone concentration measured in nmolL^{-1}/ cortisol concentration measured in nmol*L^{-1})*10^3*

The normality of data distribution was assessed with the Shapiro-Wilk test and visual analysis of plot figures. Repeated measures analysis of variance (ANOVA) was employed to analyze hormone changes during VIH, with additional time-sex interaction. The homogeneity was assessed using Levene's test. In significant main effects, post-hoc Bonferroni correction was used. The effect size was determined by partial eta squared ($p\eta^2$) and omega squared (ω^2) for significant relationships and classified as small ≤ 0.06, moderate 0.07–0.14, or large > 0.14. Moreover, the Pearson correlation coefficient has been computed to identify the relationship between hormonal changes and age, anthropometric indices, respiratory muscle strength, and training experience. A conventional approach to interpreting a correlation coefficient was applied: 0.00–0.09 was considered a negligible correlation, 0.10–0.39 was considered a weak correlation, 0.40–0.69 was considered a moderate correlation, 0.70–0.89 was considered a strong correlation, 0.90–1.00 was considered a very strong correlation (Schober, Boer, and Schwarte, 2018). Significance was set at $p<0.05$. All statistical analyses and figures were prepared with the JASP statistical package (JASP Team, Amsterdam, Netherlands, Version 0.17.2).

3. Results

There was a statistically significant effect for C changes ($F = 13.101$, $p = 0.002$, $\eta_p^2 = 0.421$, $\omega^2 = 0.08$). The C concentration was significantly lower after VIH (Mean Difference = −32.49 ± *39.13* nmol*L^{-1}). No significant effects for T and T/C ratio were observed. Also, no significant effects for any time and sex interactions were found. The result summary is presented in Table 2. As noteworthy individual variability was observed, Fig. 1 presents individual variance in pre- and post-VIH measurements with raincloud plots.

The Pearson correlation results between hormonal changes and age, anthropometric indices, respiratory muscle strength, and training experience are presented in Fig. 2. Amongst many, significant moderate correlations between the percentage of body fat and changes in C (r=-0.464, p=0.045), body mass and changes in T (r=0.516, p=0.024), height and changes in T (r=0.509, p=0.026) were found. Additionally, a moderate correlation between the percentage of body fat and changes in T was close to the traditional significance level (r=-0.439, p=0.059).

4. Discussion

The main goal of this study was to measure acute changes in C and T associated with VIH and provide initial insight into VIH's role in athletes' autoregulation and recovery practices in sports training. To our knowledge, this is one of the first studies regarding the effect of VIH on hormonal response, and the first one conducted with the participation of well-trained athletes. We observed a statistically significant effect for C changes ($p = 0.002$). The C serum concentration was significantly lower after VIH (Mean Difference = −32.49 ± *39.13* nmol*L^{-1}). Furthermore, we found no significant effects for T, T/C ratio, and any sex and time interactions. As we stipulated that VIH might lower C and only inconsiderably influence T serum concentrations, the obtained results align with the working hypotheses. Notable individual variability was observed for all monitored indices. Noteworthy correlations between the percentage of body fat and changes in C (r=-0.464, p=0.045), body mass and changes in T (r=0.516, p=0.024), height and changes in T (r=0.509, p=0.026) and the percentage of body fat and changes in T (r=-0.439, p=0.059) were observed. Contrary to the initial hypothesis, no significant correlations between hormonal changes and respiratory muscle strength were found.

The observed changes in C are aligned with long-standing research suggesting that C may decrease after efforts that are neither long nor strenuous (Davies and Few, 1973; Kaciuba-Uscilko et al., 1992). Davies and Few (1973) found a significant decrease in C was associated with intensity corresponding to around 60 % of VO_2max. Kaciuba-Uscilko et al. (1992) confirmed that the intensity of approximately 60 % of VO_2max might be the threshold for stimulating C secretion during exercise. Still, low-intensity but long-duration exercise may cause an increase in C concentration. Although the analysis of load and intensity related to RMT remains scarce, singular articles suggest that perceived effort and physiological reaction are similar to low- or moderate-intensity exercise (Kowalski et al., 2023; Iqbal et al., 2023). Therefore, similar hormonal responses were expected. Noteworthy, our study participants were instructed to employ diaphragmatic breathing patterns. The appropriate diaphragmatic breathing was shown to increase the antioxidant defense status after exhaustive exercise and was correlated with concomitant stress reduction and a decrease in C (Martarelli et al., 2011). Moreover, it was associated with vagus nerve stimulation and positively affected emotional state and well-being (Bordoni, Marelli, and Bordoni, 2016; Bordoni et al., 2018). Similarly, only 3 minutes of VIH recovery protocol significantly decreased the perceived rating-of-fatigue after anaerobic effort in well-trained short-track speedskaters (Kowalski et al., 2024). Overall, the combination of the observations mentioned before suggests that VIH may serve as an efficient recovery protocol in an athletic setting. This may be particularly useful in women, as hypercortisolism associated with exercise-induced amenorrhea exhibits a greater negative impact on health, possibly from an interaction between the hypothalamic–pituitary–adrenal axis and the hypothalamic-pituitary-gonadal axis (De Luccia, 2016). Practical application of C-lowering properties of VIH in triathletes might include stress and arousal management during pre-competition taper or on a race day, active recovery between demanding training sessions or races, and long-term well-being maintenance.

The available body of evidence regarding T responses to exercise remains weak and shows great inter-study and inter-individual variability (Hayes et al., 2015; Riachy, McKinney, and Tuvdendorj, 2020). The influence of exercise on T concentration may depend on exercise mode and population (Riachy, McKinney, and Tuvdendorj, 2020). Moreover, the T increase was often found only after resistance exercise involving large muscle mass (Migiano et al., 2010) or high-intensity endurance exercise, and no significant changes were observed after low and moderate intensity (Jezová et al. 1985). As VIH represents relatively short and low- to moderate-intensity effort with negligible resistance stimulus, limited influence on T concentration aligns with the research mentioned above. Furthermore, lack of the significant changes in T after VIH may be explained by complex and multifactorial mechanisms that may influence T secretion, disposal, and consequently concentration (Wang and Zhao, 2023). Aside from VIH-induced changes other physiological, psychological, and environmental factors like nutrition, sleep, emotional state, and mental fatigue may also affect T

Table 2
Changes in cortisol, testosterone, and testosterone-to-cortisol ratio before and after Voluntary Isocapnic Hyperpnoea.

Variable	Females (n=7)		Males (n=12)		p-value
	Before	After	Before	After	
Cortisol (nmol*L^{-1})	428.4 ± 254.8	380.9 ± 269.2	335.5 ± 99.3	311.8 ± 106.2	0.002
Testosterone (nmol*L^{-1})	0.91 ± 0.28	0.87 ± 0.32	15.6 ± 6.2	16.0 ± 2.1	0.526
Testosterone to cortisol ratio	2.1 ± 1.1	2.3 ± 1.3	42.9 ± 25.6	49.8 ± 33.0	0.109

Values are mean ± standard deviation.

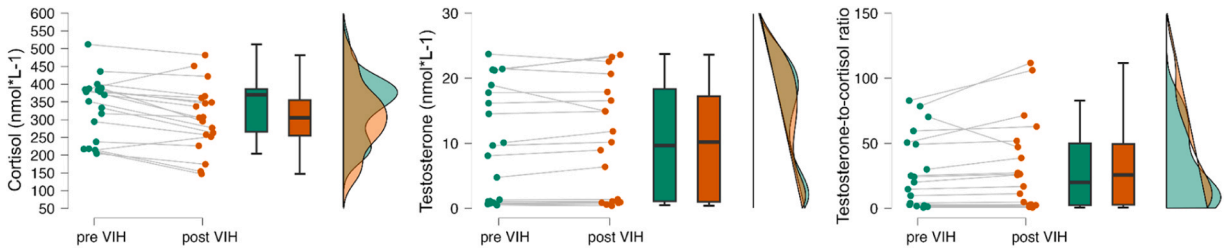

Fig. 1. Individual variance in pre- and post-VIH measurements. VIH - Voluntary Isocapnic Hyperpnoea.

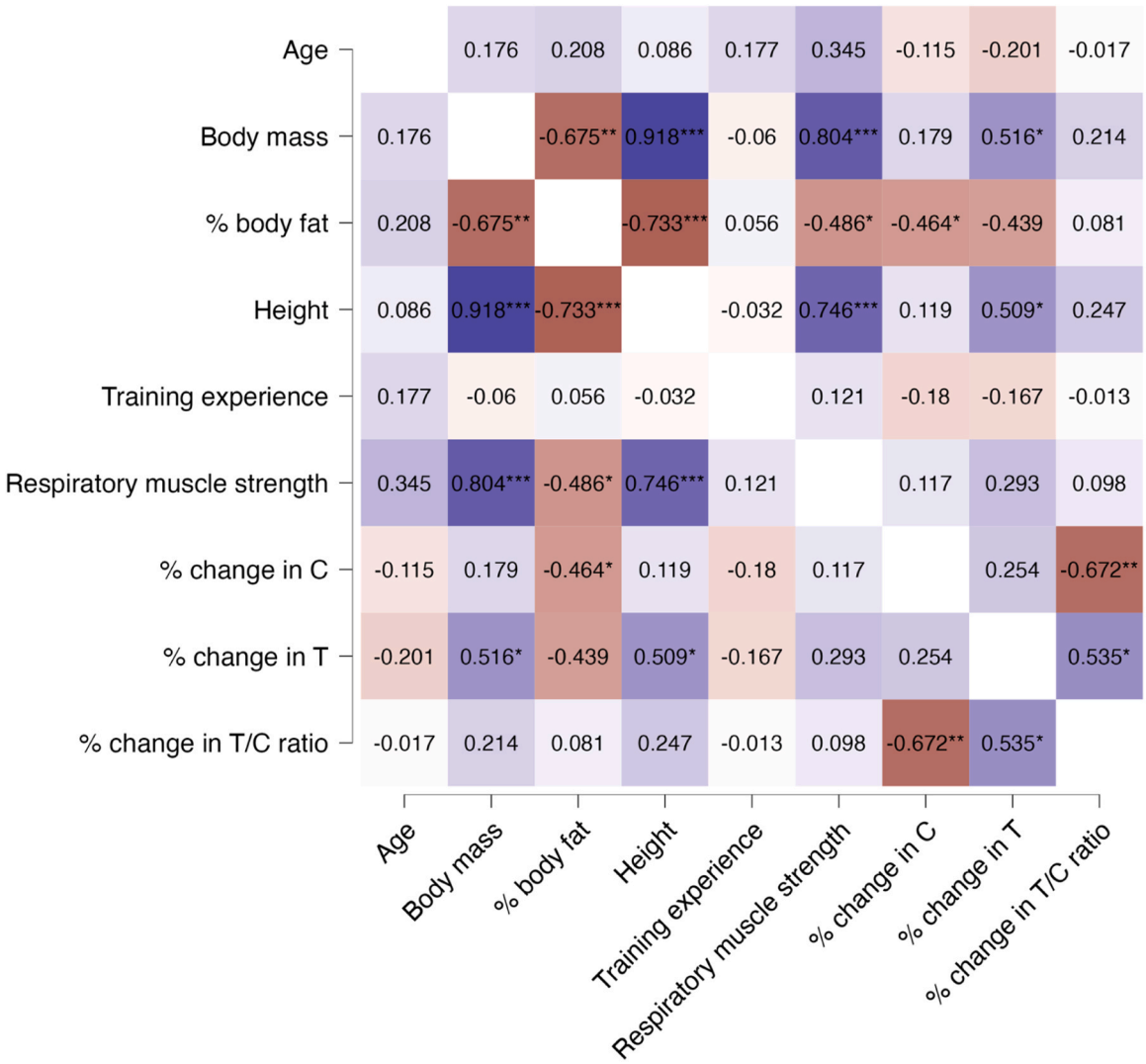

Fig. 2. Heatmap of correlations between hormonal changes and age, anthropometric indices, respiratory muscle strength, and years of training experience. C - serum cortisol, T - serum testosterone. Significant correlations were flagged as follows: *p <0.05, **p<0.01, ***p<0.001.

concentration (Mathur and D'Cruz, 2011; Ambroży et al., 2021; Zurek et al., 2022).

Interestingly, the hormonal response to VIH or hyperventilation may depend on familiarisation with the protocol level and training status. Djarova et al. (1986) investigated healthy males, found an increase in C after only 3 minutes of hyperventilation, and associated the respiratory effort with the acute stress response (Djarova et al., 1986). Similarly, Kowalski et al. (2023) in a 6-week study of different RMT methods in well-trained athletes presented mixed results for hormonal changes during the first measurements. However, they observed a positive hormonal response during measurements in the last week of the intervention (Kowalski et al., 2023). Except for the different RMT experience, both groups presumably differed in training status and ventilatory efficiency, as both vary in well-trained endurance athletes compared to the typical healthy population (Petek et al., 2022; Kasiak et al., 2024; Srivastava et al., 2024). Noteworthy, the practical effectiveness of the intervention may be associated with between-gender differences in perceived stress (Dawson et al., 2014). Gender, experience with VIH, and fitness level may play a crucial role in the investigated protocol's effectiveness and should be considered by researchers, practitioners, and athletes. Therefore, the application of RMT-derived protocols might require a different approach in patients compared to

well-trained athletes.

The study design is associated with the exploratory character of our investigation and is not free from limitations. The noteworthy limitation is the lack of a control group or within-group control measurement session. The study investigated a homogeneous group of well-trained triathletes, therefore our findings should not be extrapolated to different populations. Moreover, we performed only single pre- and post-VIH measurements. As a consequence, the individual variation in hormone secretion may have influenced the obtained results. In the presented study we did not include the menstrual cycle phase in the analysis, as we measured only the acute effects of RMT within not only the same day but within the same hour. Therefore, the possible differences between different menstrual cycle phases were outside the scope of the study. Finally, although the required sample size was met, the study would have benefited from a larger number of participants. The practical application of RMT protocols in the context of hormonal balance in athletes requires further investigation. Future research might employ more measurement time points to assess the individual variability of hormonal secretion after VIH or use different RMT techniques and protocols. Our correlation analysis indicated a possible association of VIH effectiveness on C and T with the percentage of body fat (speculatively - fat-free mass as well). Therefore future investigation should include this angle in analyzing factors possibly affecting the effectiveness of VIH regarding hormone secretion. Moreover, an investigation regarding the influence of hormonal contraception and the menstrual cycle phase might provide valuable insights into further understanding of the studied issue. Finally, the hormonal balance should be analyzed in the context of real-world stress, anxiety, arousal perception, athletic performance, and response to exercise.

The presented study provides initial insight into VIH's role in athletes' hormonal balance to possibly guide exercise prescription, autoregulation, emotional state management, and regeneration practices in sports training and physical therapy. The C-lowering properties of VIH exhibit the potential for optimizing athletes' recovery strategies.

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CRediT authorship contribution statement

Andrzej Klusiewicz: Writing – review & editing, Writing – original draft, Resources, Project administration, Investigation, Conceptualization. **Wojciech Waleriańczyk:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Tomasz Kowalski:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zbigniew Obmiński:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization.

Data availability

Data will be made available on request.

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